

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
Name	Kishor Sreenivasan	Register No.	3122235001068
Due Date	30/3/2026		

Experiment 6:
Decision Tree and Random Forest: A Comparative Classification Study

1. Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2. Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

3. Theory

Decision Tree Classifier

A Decision Tree is a supervised learning model that recursively splits the feature space using impurity measures to form decision rules. Important concepts include Gini Index, Entropy, tree depth, and node splitting. A major limitation of a single Decision Tree is that it can easily overfit the training data when the tree becomes too deep.

Random Forest Classifier

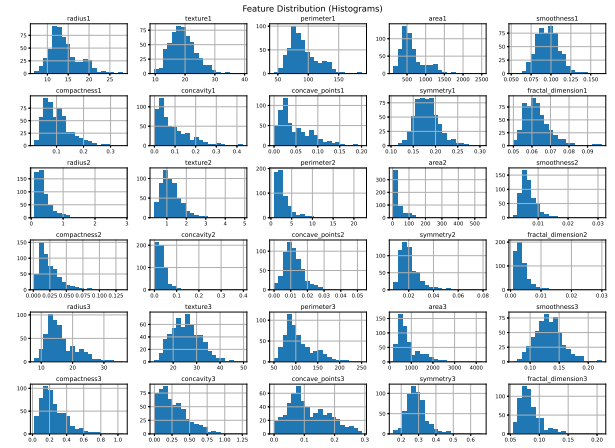
Random Forest is an ensemble learning technique that combines multiple decision trees trained on bootstrapped samples of the data. It uses random feature selection and majority voting to

improve predictive performance. Compared to a single tree, Random Forest usually offers better generalization and lower variance.

4. Implementation Details

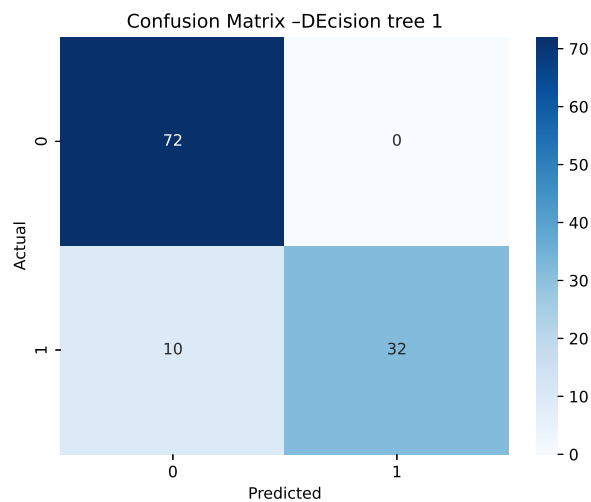
- Loaded the dataset and encoded the class labels.
- Performed exploratory data analysis using class distribution and feature correlation analysis.
- Split the dataset into training and testing sets using an 80–20 ratio.
- Trained a Decision Tree classifier.
- Defined a hyperparameter search space for Decision Tree and evaluated it using 5-Fold Cross-Validation.
- Selected the best Decision Tree hyperparameters based on average cross-validation performance.
- Trained a Random Forest classifier.
- Repeated hyperparameter tuning for Random Forest using 5-Fold Cross-Validation.
- Compared both models using accuracy, precision, recall, F1-score, confusion matrix, ROC curve, and AUC.

5. Visualizations

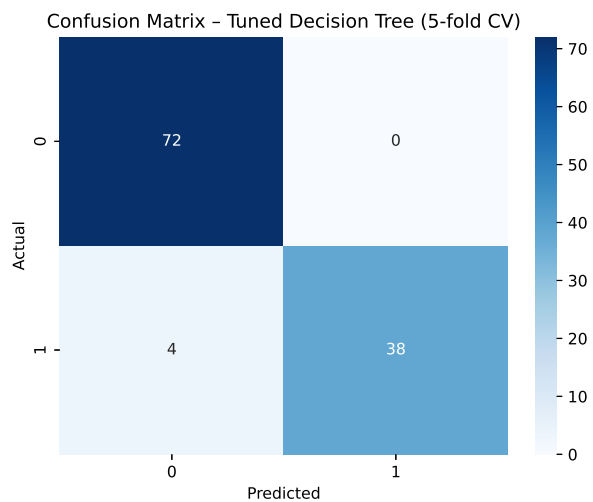


(a) correlation heatmap

(b) Feature distribution

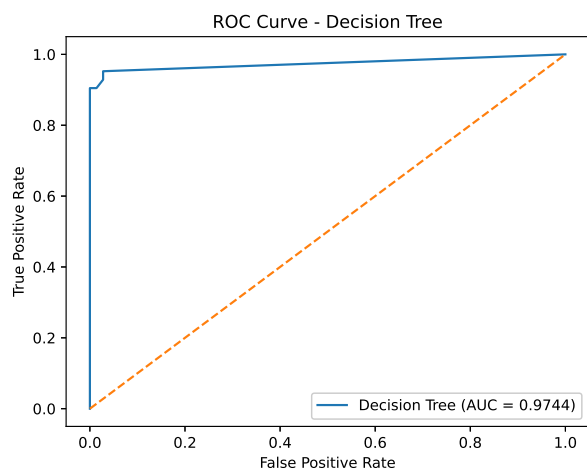


(a) Base Decision Tree confusion matrix

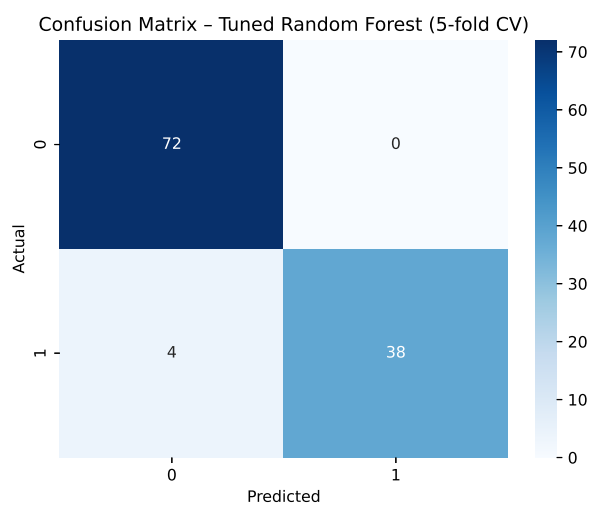


(b) Tuned Decision tree confusion matrix

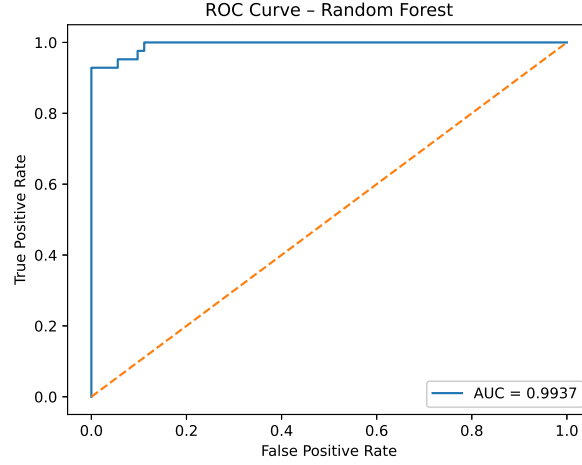
Figure 2: Decision Tree



(a) Decision Tree ROC curve



(b) TunedRandom forest confusion matrix



(a) Random Forest ROC curve

6. Hyperparameter Tuning Results

Decision Tree Cross-Fold Results(Best accuracy)

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
entropy	5	0.9362	0.95

Table 1: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Random Forest Cross-Fold Results

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
200	5	sqrt	0.9714	0.95

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

5-Fold Cross-Validation Performance Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9011	0.9560	0.9121	0.9560	0.9560	0.9362
Random Forest	0.9560	1.0	0.9450	0.9670	0.9890	0.9714

Table 3: 5-Fold Cross-Validation Accuracy Comparison

7. Evaluation Metrics

Model	Accuracy	Precision	Recall	F1-score	AUC
Decision Tree	0.9362	1.00	0.90	0.95	0.9744
Random Forest	0.9714	1.00	0.90	0.95	0.9937

Table 4: Model Evaluation Metrics

8. Observation Questions and Answers

1. How does tree depth affect overfitting in Decision Trees?

Tree depth directly controls the complexity of the model. When the tree depth is very small, the model may underfit because it cannot capture important patterns in the data. As tree depth increases, the Decision Tree can fit the training data more closely. However, if the depth becomes too large, the model starts learning noise and small fluctuations from the training set, which leads to overfitting. This reduces its ability to generalize to unseen test data. Therefore, controlling depth is important to maintain a balance between bias and variance.

2. Which hyperparameter had the greatest impact on performance?

Among the explored hyperparameters, **max depth** usually has the greatest impact on the performance of both Decision Tree and Random Forest models. In a Decision Tree, depth determines how specific the splitting rules become, directly affecting underfitting or overfitting. In Random Forest, **n_estimators** also plays an important role because increasing the number of trees generally improves stability and reduces variance. However, **max depth** remains the most influential factor because it strongly affects model complexity and generalization.

3. How does Random Forest improve generalization?

Random Forest improves generalization by combining multiple Decision Trees instead of relying on a single tree. Each tree is trained on a different bootstrapped sample of the dataset, and each split considers only a random subset of features. This introduces diversity among the trees. When their predictions are combined using majority voting, errors from individual trees tend to cancel out. As a result, Random Forest reduces variance, becomes less sensitive to noise, and usually performs better on unseen data than a single Decision Tree.

4. Did ensemble learning always improve performance? Why or why not?

Ensemble learning usually improves performance, but it does not guarantee improvement in every case. Random Forest often performs better than a single Decision Tree because it reduces overfitting and improves stability. However, the amount of improvement depends on factors such as dataset size, feature quality, hyperparameter settings, and the strength of the individual trees. If the base

model is already well-tuned or if the dataset is simple, the performance gain may be small. In some cases, poor hyperparameter choices or insufficient diversity among trees can limit the advantage of ensemble learning.

9. Learning Outcomes

- Learned how a Decision Tree classifier works using recursive splitting.
- Understood the role of impurity measures such as Gini Index and Entropy.
- Observed how tree depth influences underfitting and overfitting.
- Studied how Random Forest reduces variance through bagging and random feature selection.
- Performed 5-Fold Cross-Validation for robust hyperparameter tuning.
- Compared single-tree and ensemble-tree models using standard classification metrics.

10. Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-Fold Cross-Validation. Hyperparameters were selected based on average cross-validation performance to improve generalization. A single Decision Tree is simple and interpretable but more prone to overfitting. Random Forest overcomes this limitation by combining multiple trees, thereby reducing variance and improving stability. Overall, Random Forest is generally expected to provide better generalization performance than a single Decision Tree.